

Hei Shing Helson Go

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SUMMARY

Ph.D. researcher in autonomy and intelligent control with seven years designing, proving, and validating algorithms for autonomous robots. Deep background in system modeling, optimization and machine learning, with experience spanning perception, localization, planning, and control. Proficient in sim-to-real development, with a track record of end-to-end deployment of state-of-the-art algorithms on physical autonomous vehicles, including under competitive, time-constrained conditions. Peer-reviewed publications at top robotics venues (ICRA); fluent in Python, PyTorch, performance-critical **C++**, and **Rust**.

RESEARCH EXPERIENCE

University of Toronto Institute for Aerospace Studies

Sep 2020 – Oct 2025 | Toronto, ON

Ph.D. Researcher — Flight Systems & Control Lab

Doctoral research: optimal and learning-based localization and control for autonomous vehicles

- **Created** observability-aware planning and control algorithms to safeguard sub-0.5 m positioning confidence without GPS for teams of mutually sensing autonomous vehicles.
- **Established** a new mathematical link between system observability and achievable localization precision, translating it into an active sensing principle that shapes vehicle behavior to maximize the information from sparse sensor measurements, demonstrating **over 75%** improvement in positioning confidence versus baseline trajectories in Monte Carlo studies.
- **Introduced** autodiff tooling (*JAX*) from machine learning into classical optimal control, using automatic differentiation to evaluate complex pushforwards that make up on-manifold Lie derivatives within nonlinear-observability metrics for optimization.
- **Benchmarked** a learned observability model trained on synthetic data against an analytical metric, finding that autodiff on a known system makes the analytical metric fast enough and more accurate.
- **Validated** the methods end-to-end in high-fidelity, closed-loop simulation and on physical autonomous vehicles with inter-drone trackers, reducing real-world localization uncertainty by **54–61%**, and published the results across two peer-reviewed robotics and control venues (See Selected Publications).

Applanix (Trimble) collaboration: LiDAR-inertial odometry for GNSS-denied localization

- **Re-engineered** a modern C++20 LiDAR-inertial odometry library based on AriseLab *Super-Odometry* to complement Applanix's APX-15 GNSS system for navigating drones in GNSS-denied environments.
- **Implemented** a modular perception pipeline: edge/planar feature extraction, IMU preintegration, scan-to-map registration (ICP), an iteratively-reweighted Gauss-Newton optimization that replaced *Ceres Solver*, and an octree local map that replaced *iKd-Tree*.
- **Flight-tested** the system in closed loop: streamed a **200 Hz** IMU-propagated odometry feed into *PX4*'s EKF2 and flew a drone in position-controlled mode on the LiDAR odometry as its sole position source; covered **248 m** autonomously at up to **6 m/s** without triggering EKF2 warnings.

Drone Delivery Canada collaboration: data-driven behavior and dynamics modeling

- **Designed and validated** a data-driven control system for a slung load-transporter drone, which learns the load's dynamics from flight data using the **gptorch** framework, to keep the payload stable under disturbance, and demonstrated it end-to-end in a live, real-world delivery mission.

APPLIED & LEADERSHIP EXPERIENCE

Early-Stage Autonomous Drone Racing Venture

Oct 2025 – Present | Toronto, ON

Co-founder & Chief Engineer

- **Re-implemented** the University of Toronto Autonomous Drone Racing Team's autopilot as embedded real-time control firmware in Rust (**no_std**, ARM Cortex-M7) while exploring commercialization of the platform, aiming to create a lightweight 'autonomy box' hardware for robots that encapsulates nonlinear model-predictive control, time-optimal path planning, incremental nonlinear control with online, data-driven learning of control effectiveness.

- **Transitioned** the racing autopilot’s Error-State Kalman Filter (ESKF) to a multi-sensor navigation system fusing IMU, barometer, magnetometer, GNSS, and motion-capture measurements; implemented fault-tolerant sensor supervisors for safe outdoor operation; validated it against reference implementations via equivalence tests and benchmarks.

University of Toronto Autonomous Drone Racing Team

Sep 2024 – 2025 | Toronto, ON

Engineering Lead

- **Led** a team of 7 that developed an optimization- and machine-learning-based autopilot for a high-speed autonomous drone, setting technical direction across modeling, development, and closed-loop testing, placing **6th worldwide** at the 2025 Abu Dhabi Autonomous Racing League.

Samsung AI Center

Jun 2019 – Aug 2019 | Montreal, QC

Intern — Machine Learning Research Project Support

- **Translated** a novel machine-learning resource-allocation algorithm into a testable prototype and built a large-scale simulation to verify it in an automated testing pipeline.

AERO McGill

Sep 2016 – May 2019 | Montreal, QC

Technical Lead

- **Led** a student team of 6 to design and build an autonomous aerial system that won **2nd place** and the **Judges’ Award for Professionalism** at the 2019 Unmanned Systems Canada national competition.

OPEN-SOURCE PROJECTS

- **isaac-drone-racing**: RL drone racing in NVIDIA Isaac Sim (PPO, sim-to-sim transfer, perception-aware reward shaping).
- **discrete_pid**: published `no_std` Rust PID crate, Simulink-verified, with tests and benchmarks.

EDUCATION

University of Toronto

2019 – 2025 | Toronto, ON

Ph.D., Aerospace Science & Engineering

Dissertation: “Cooperative Navigation and Control of UAVs in GNSS-Denied Environments.”

McGill University

2015 – 2019 | Montreal, QC

B.Eng., Mechanical Engineering

TECHNICAL SKILLS

Programming: Python, performance-critical C++ (modern C++17/20), Rust (real-time embedded, `no_std`), C, Kotlin; PyTorch, JAX, CasADi, Eigen, nalgabra, OpenCV, CMake, pybind11; Linux, Git

Machine Learning & AI: deep learning (PyTorch, JAX), data-driven / imitation-style learning of dynamics, statistical modeling, numerical and convex optimization

Robotics & Autonomy: perception, localization and mapping, sensor fusion and estimation, motion planning, nonlinear / model-predictive control, behavior and dynamics modeling, LiDAR

Math Foundations: linear algebra, calculus, probability and statistics, estimation theory, optimization

Methods: first-principles modeling, test-driven development, automated testing pipelines, high-fidelity simulation, hardware-in-the-loop and field validation

SELECTED PUBLICATIONS

- [1] **H. S. Helson Go**, C. L. Chong, L. Qian, and H. H.-T. Liu. “Observability-Aware Control for Quadrotor Formation Flight with Range-only Measurement.” *AIAA J. of Guidance, Control, and Dynamics (JGCD)*, 2026.
- [2] **H. S. Helson Go** and H. H.-T. Liu. “Trajectory Optimization for Cooperatively Localizing Quadrotor UAVs.” *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2024.
- [3] **H. S. Helson Go**, L. Qian, and H. H.-T. Liu. “Data-Driven and Robust Path-following Control of a Quadrotor Slung Load Transport System.” *AIAA SciTech Forum*, 2023.